# Formalizing AI Scientist Systems: A Component Framework and Theoretical Analysis

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**Post-submission revised manuscript — not the version of record.** This manuscript was refined after the CAISc 2026 submission deadline and is *not* the paper that was reviewed and archived. The version of record is the camera-ready deposited on OpenReview; nothing here alters it. Every revision made since that deposit is listed, with its rationale, in Appendix D. For bibliographic identity, cite the version of record; when relying on a claim corrected here, cite or identify this dated revision alongside it and do not attribute the corrected claim to the archival text. The correction should also be linked from the official record when that repository permits.

**Revised:** 2026-07-19; last revised 2026-07-22.

## Abstract

Existing surveys organize AI Scientist systems by capabilities and workflow stages. We propose Servo $(S, G, E, V, M, \pi)$ as a non-exhaustive, source-traceable diagnostic schema and vocabulary for recording component interfaces, evaluation events, versioned artifacts, routed edges, memory, and policy. The six-component core comprises the **S**earch space, hypothesis **G**enerator, **E**xperiment executor, **V**alidator, **M**emory, and search **P**olicy ($\pi$); the environment observation law $O_{\text{env}}$, artifact state $A$, and optional artifact synthesis or revision interface $W_A$ are explicit boundary objects rather than additional core components. We apply the schema to six versioned cases from five lineages and seven selected, frozen domain anchors. It replaces a single closure label with four conservative, set-valued predicates whose statuses require source-traceable witnesses within one case, version, configuration, and task regime. The case and anchor records illustrate use of the vocabulary; they are not a representative sample, prevalence estimate, causal test, or demonstration that Servo is superior to another framework. Appendix A uses standard Bayesian results only to delimit observation, validation, and novelty in fixed idealized settings. We identify three open problems as prospective design questions: independently validated automated novelty assessment, integration of information-seeking policies, and multi-fidelity integration.

## 1 Introduction

AI Scientist systems—agents that generate hypotheses, execute or coordinate experiments, and synthesize knowledge with varying human authority—have proliferated rapidly [Boiko et al., 2023, Lu et al., 2024, 2026, Feng et al., 2026, Liu et al., 2026]. Capability- and stage-based surveys provide shared organizing frameworks [Wei et al., 2025, Tie et al., 2025], but different analytical purposes foreground different system boundaries. The questions pursued here are narrower: *Which evaluation event is connected to which later action? Which artifact version carries that connection? Which kind of feedback, if any, is established within a bounded reported configuration? What evidence would support novelty assessment or information-seeking experimental policy?*

Published at 1st Conference For AI Scientists (CAISc 2026).

We propose Servo[1] $(S, G, E, V, M, \pi)$, organized using selected POMDP and Bayesian Experimental Design terminology (Section 4). We apply the schema to six versioned cases from five lineages and separately describe seven selected source anchors (Sections 5, 6). The contribution is the explicit contract and its source-traceable application, not a complete ontology, causal theory, population taxonomy, or performance ranking.

**Scope.** Servo is a non-exhaustive system-level diagnostic schema. Each case is a versioned system configuration in a bounded task regime, identified by case, lineage, version, configuration, and task-regime identifiers. Inclusion records source-traceable variation in execution, evaluation, artifacts, routing, policy, authority, or fidelity; it is not a claim that the cases exhaust or represent AI Scientist systems.

## 2 Background

Following Kaelbling et al. [1998], we borrow two organizing concepts from POMDPs: a policy maps information about an incompletely observed environment to an action, and a belief state summarizes that information. A fully instantiated POMDP would require a world-state space $\mathcal{X}$, action space $\mathcal{A}$, transition kernel, observation kernel, reward function, initial belief, and discount. Servo does not specify those objects and is not a POMDP instantiation. Its $S$ is a candidate search space rather than the world state, and any $b_t$ is used only when a bounded source defines a belief or posterior. Appendix A separately analyzes fixed statistical models. Posterior concentration is not any of Servo's four closure predicates, and no predicate status defines a trustworthy scientific outcome. An outcome study must predeclare a gate-independent endpoint $Y_q$ for each target property $q$. Bayesian Experimental Design [Chaloner and Verdinelli, 1995, Rainforth et al., 2024] provides one possible normative information criterion for $\pi$: select $d^*$ maximizing Expected Information Gain (EIG),

$$d^* = \arg\max_{d \in \mathcal{D}} \mathrm{EIG}(d) = \arg\max_{d} \mathbb{E}\left[H(\theta) - H(\theta \mid y, d)\right] \tag{1}$$

where $H(\theta)$ is prior entropy over parameters $\theta$ and $H(\theta \mid y, d)$ is expected posterior entropy after observing $y$ under $d$. EIG can be approximated via variational methods [Foster et al., 2019], amortized policies [Foster et al., 2021], and RL [Blau et al., 2022]. Robot Scientist [King et al., 2004] and GNoME [Merchant et al., 2023] illustrate domain-specific active-learning policies; the bounded LLM case sources used here do not state an EIG objective.

## 3 Related Work

**AI Scientist systems and direct survey frameworks.** Individual systems have been characterized in detail [Lu et al., 2024, 2026, Schmidgall et al., 2025, Boiko et al., 2023], while recent surveys provide shared comparison frameworks. Wei et al. [2025] integrate capabilities and workflow, and Tie et al. [2025] organize systems by methodological stages and abstraction layers. Servo instead foregrounds versioned case identity, events, artifacts, routes, and predicate witnesses. This is a difference in analytical unit, not evidence that Servo is more complete or useful.

**Autonomous-science frameworks.** Automated feedback architectures predate the LLM era: the Robot Scientist [Sparkes et al., 2010] integrated hypothesis generation, robotic experimentation, statistical testing, model update, and active experiment selection [King et al., 2004]. More recent architectures include InternAgent (formerly NovelSeek) [InternAgent Team, Shanghai Artificial Intelligence Laboratory, 2025] and Co-Scientist [Gottweis et al., 2026]. We therefore position Servo not as the first unified framework but as a vocabulary for separating observation generation, evaluation events, artifacts, routed edges, and acceptance. Its diagnostic value is illustrated, not independently tested against another framework.

**Decision-theoretic foundations.** The POMDP framework [Kaelbling et al., 1998] and BED [Chaloner and Verdinelli, 1995, Rainforth et al., 2024] are classical, with a long history of Bayesian-optimal experiment selection in automated science—most directly the Robot Scientist's

[1] The servomechanism metaphor motivates tracing a feedback path; it does not imply that every recorded path is reliable, epistemic, autonomous, or scientifically successful.

active-learning policy [King et al., 2004]. Servo borrows terminology rather than claiming a complete POMDP or priority for BED. Its machine-readable contract records validation target, evidence, evaluator, decision role, artifact state, and routed edge without treating those fields as a reliability hierarchy.

**Formal models and artifact infrastructure.** Taniguchi et al. [2025] model collective scientific inference, whereas Servo records one system-level case and deliberately compresses social organization. Liu et al. [2026] foreground artifact format, provenance, and verification. The Servo event–artifact schema can record such artifacts as versioned $A$ states and, where the source reports it, an artifact synthesis or revision interface $W_A$; this is a boundary mapping rather than subsumption.

**Provenance and workflow-run standards.** PROV-DM already defines entities, activities, agents, generation, usage, derivation, revision, and association [Moreau and Missier, 2013]. Workflow Run RO-Crate similarly records a workflow plan and run with code and input/output artifacts [Workflow Run RO-Crate Community, 2023]. Servo does not redefine these base provenance objects. Its artifacts map to PROV entities, events to activities or run actions, producers and consumers to generation and usage, and human or system actors to agents and associations. The added profile is narrower: scientific target properties, validator-reliability records, typed epistemic and control routes, bounded case identity, component-specific authority, and predicate-specific closure witnesses. Table 1 states this reuse boundary; formal conformance would additionally require an executable serializer and the relevant conformance tests.

| Servo record | Existing analogue | Servo-specific scientific semantics |
|---|---|---|
| Artifact/version | PROV Entity; derivation/revision | Version identity required by a bounded predicate witness |
| Event | PROV Activity; RO-Crate run action | Target property, evaluator, reliability evidence, update semantics |
| Actor/route | Agent/Association; generation/usage | Component authority and typed epistemic, control, revision, or mediation role |
| Closure witness | Connected provenance subgraph | Minimum predicate path and evidential decision status |

Table 1: Construct-level provenance crosswalk. Servo is a scientific-diagnostic profile over reusable provenance concepts, not a replacement base ontology.

| Framework | Unit and aim | Foregrounded resolution | Deliberate omission / tradeoff | Source |
|---|---|---|---|---|
| Servo | System-level discovery loop comparison | Components; property-specific validator channels; operational versus terminal feedback | Compresses agent communication and social organization; not a complete generative model | This work |
| Wei et al. | Capability–workflow–domain synthesis | Five foundational capabilities and four dynamic discovery stages | Validator-channel identity and operational versus terminal routing are not the organizing unit | Wei et al. [2025] |
| Tie et al. | Six-stage methodology by four abstraction layers | Stage coverage across domains, tasks, architectures, and capabilities for 50+ systems | Does not separately encode memory, policy, or complete validator feedback paths | Tie et al. [2025] |

Table 2: Direct construct-level comparison with AI Scientist survey frameworks, using each framework's native unit. This is not a performance ranking. Servo's claimed increment is selected system-level channel and interface resolution, not the first unified framework or information that the alternatives are incapable of representing.

POMDP, InternAgent, CPC-MS, and ARA remain complementary rather than direct survey-taxonomy baselines. POMDP supplies a sequential decision model when states and kernels are specified [Kaelbling et al., 1998]; InternAgent foregrounds implemented multi-agent organization [InternAgent Team, Shanghai Artificial Intelligence Laboratory, 2025]; CPC-MS models community-level epistemic dynamics [Taniguchi et al., 2025]; and ARA foregrounds artifact provenance and replay [Liu et al., 2026]. Servo does not subsume these analytical units.

## 4 The Servo Framework

We define an AI Scientist system as a tuple $(S, G, E, V, M, \pi)$ with the following components:

$$\text{Servo} = (S,\ G,\ E,\ V,\ M,\ \pi) \tag{2}$$

$S$**: Search Space** The currently represented candidate hypotheses, programs, molecules, experimental conditions, or other objects subject to exploration. $S_t$ may be fixed or expanded;

calling a process open-ended requires an explicit expansion rather than merely searching a large fixed set.

$G : M \times S \to (\mathcal{C}, \Delta S)$**: Hypothesis Generator** A function that proposes both newly instantiated candidates and an optional search-space expansion. We set $S_t^+ = \text{expand}(S_t, \Delta S_t)$ and require $\mathcal{C}_t \subseteq S_t^+$. Fixed-space search has $\Delta S_t = \varnothing$; progressive widening or a newly introduced representation must be recorded in $\Delta S_t$. $G$ supplies options and expansions but does not select the next system action.

$E \equiv E_{\text{exec}} : h \times d \times P \times f \to u$**: Experiment Executor** The system-controlled component that turns a selected hypothesis, design, protocol, and fidelity into an executed intervention or computational action $u$. It may be a code runner, simulator, robot, or protocol orchestrator, with fidelity ranging from computational proxy to physical wet-lab execution. The stochastic response of nature or a simulator is not the executor itself. We denote that auxiliary environment and measurement law by $O_{\text{env}}(\mathrm{d}o \mid x, u)$, where $x$ is an environment state rather than a Servo component. Hence an observation is drawn as $o \sim O_{\text{env}}(\cdot \mid x, E_{\text{exec}}(h, d, P, f))$, with a Dirac law as the deterministic special case. $O_{\text{env}}$ is not a seventh Servo component or a field inferred by the architectural coding; it must be supplied by a statistical model before a belief update or EIG is computable. $V$ remains a separate contextual assessment of the realized evidence.

$V = \{V_j : (z_j, A_v) \to r_j\}_{j \in \mathcal{J}}$**: Validator events** A family of context-dependent assessment functions rather than one compulsory post-observation call. $A_v$ identifies every versioned artifact consumed by the event; its producer, version, and case boundary must resolve in the artifact records. The contextual input $z_j = (h, d, P, o, M, K_{\text{ext}}, \tau)$ may additionally contain the candidate claim or hypothesis, design, protocol, observation, internal memory, an external reference corpus, and an evaluation-time cutoff. Its result $r_j$ may contain property-specific components for executability, empirical adequacy, reproducibility, novelty, significance, and acceptance. A particular validator uses only the coordinates it needs; for example, an execution check may depend only on $(P, o)$, whereas field novelty necessarily compares $(h, o)$ with time-indexed external prior art $(K_{\text{ext}}, \tau)$. Validators are coded by separate facets rather than additive layers: target property (e.g., executability, empirical adequacy, novelty, significance), evidence source (execution trace, measurement, replication, literature, review), evaluator substrate (program, statistical model, LLM, human, hybrid), decision role (diagnostic, search control, or acceptance), and external independence (internal, shared, or independent). Multiple values may apply within a facet; values from different facets must not be added or interpreted as maturity levels. An event does not create a route by itself. A separately evidenced edge may route $r_j$ to $D_j \subseteq \{S, G, E, V, M, \pi, A, W_A, \text{external}\}$. Artifact revision is represented specifically as a route to $W_A$ followed by an `artifact_revision` edge that produces a new artifact version $A_{v+1}$; mutating $A_v$ in place is not an admissible shortcut.

$M$**: Memory** Structured as $(M_e, M_s, M_p)$. $M_e$ (episodic) stores individual records $(h_i, o_i, r_i)$; $M_s$ (semantic) stores distilled knowledge patterns; $M_p$ (procedural) stores learned protocols. The structure of $M$ directly determines the quality of $G$ and $\pi$.

$\pi : b \times M \times \mathcal{C} \times B \times R_{\text{pre}} \to a$**: Search Policy** The decision rule that selects an action $a = (h, d, P, f)$, comprising a candidate $h \in \mathcal{C}$, an experimental design $d$, an executable protocol $P$, and a fidelity $f$. $B$ is the remaining resource budget and $R_{\text{pre}}$ contains validation signals routed before action selection. The belief state $b$ is updated after each experiment. A myopic BED policy holds the candidate and feasible-action construction fixed and selects $a$ (or its design component $d$) to maximize conditional EIG; this is a one-step action rule, not a globally optimal sequential policy. In practice, systems use reactive, greedy, active-learning, or tree-search policies. Thus $G$ generates candidates while $\pi$ chooses what to do with them, consistent with the POMDP convention that a policy maps a belief state to an action [Kaelbling et al., 1998].

**Operational event–artifact graph.** For a bounded case $c$, Servo records endpoints, versioned artifacts, evaluation events, and directed edges,

$$\mathcal{G}_c = (\mathcal{N}_c, \mathcal{A}_c, \mathcal{E}_c, \mathcal{R}_c), \tag{3}$$

where $\mathcal{N}_c$ are typed component or boundary endpoints, $\mathcal{A}_c$ are artifacts with producers and versions, $\mathcal{E}_c$ are evaluation occurrences, and $\mathcal{R}_c$ are observation, epistemic-update, feedback-control, artifact-revision, or external-only edges. Each event class has an allowed actor component, and each edge type has allowed source–destination component pairs; the package rejects graphs that violate these tuple–graph consistency constraints. Human, system, mixed, or unreported mediation is an actor facet rather than a distinct edge topology. Events and reliability evaluations are different records: an evaluation study is not a runtime event and cannot serve as a feedback edge. $A$ denotes artifact state consumed or produced by events, while the optional $W_A$ denotes a reported artifact synthesis or revision interface. Both lie at the six-component boundary; neither silently expands the core tuple. A terminal event has no outgoing internal edge. Pre-action filtering, a fixed schedule, append-only memory, formatting or copying, retry without epistemic update, and terminal assessment alone are prohibited shortcuts rather than closure witnesses.

The admissibility maps from graph records to tuple or boundary components are explicit. For event class $k$ and edge type $r$, let $\eta(k)$ be the allowed actor-component set and $\tau(r)$ the allowed ordered component pair set:

$$\begin{aligned}
\eta(\text{runtime validation}) &= \{V\}, & \eta(\text{artifact assessment}) &= \{V\},\\
\eta(\text{generation}) &= \{G\}, & \eta(\text{execution}) &= \{E\},\\
\eta(\text{artifact production}) &= \{W_A\}, & \eta(\text{state update}) &= \{M\},\\
\eta(\text{human feedback}) &= \{\text{external}\}, & \eta(\text{external assessment}) &= \{\text{external}\},
\end{aligned} \tag{4}$$

and

$$\begin{aligned}
\tau(\text{observation}) &= \{(E, V)\},\\
\tau(\text{epistemic update}) &= \{(V, M)\},\\
\tau(\text{artifact revision}) &= \{(V, W_A), (\pi, W_A)\},\\
\tau(\text{external only}) &= \{(\text{external}, \text{external})\},\\
\tau(\text{feedback control}) &= \{(V, G), (V, \pi), (M, G), (M, \pi),\\
&\quad (\pi, G), (\pi, E), (G, E), (W_A, E),\\
&\quad (\text{external}, W_A)\}.
\end{aligned} \tag{5}$$

For every event, the actor component must belong to $\eta$ of its event class; for every edge, the ordered endpoint-component pair must belong to $\tau$ of its edge type. These are integrity constraints, not an assertion that every component invocation is observable in a bounded source; unreported mappings remain evidentially unknown.

**Set-valued closure description.** Servo does not assign one yes/partial/full loop label. For each case it reports a status in $\{\text{established}, \text{not established}, \text{unknown}, \text{not applicable}\}$ for each of four predicates: *execution repair*, *experimental adaptation*, *artifact revision*, *discovery-cycle feedback*. An established predicate requires an ordered witness whose events, edges, endpoints, and artifacts remain within the same case, version, configuration, and task regime. Execution repair requires runtime failure evidence, artifact revision through $W_A$, and a later execution; experimental adaptation requires evaluation evidence to control a changed experimental action that is subsequently executed; artifact revision requires a successor versioned artifact; and discovery-cycle feedback further requires an explicit epistemic update, an intervening epistemic action and execution, and new evidence. The predicates may share records only when the path independently satisfies each contract. Human mediation is reported separately as an actor facet and may qualify any of these paths; it is not itself evidence that a loop closes.

The evidential statuses are not interchangeable. *Established* requires a positive witness. *Unknown* means that a required event, edge, order, artifact identity, or source link is missing or ambiguous. *Not established* is reserved for explicit contrary evidence or a completely reported eligible trace that fails a required transition; source silence alone is *unknown*. *Not applicable* requires a predeclared out-of-scope reason. These graph patterns, mismatch rules, and counterexamples are specified in `predicate_contract.md` and enforced by the package validator.

Human involvement is recorded as a categorical authority vector rather than the former scalar $H \in [0, 1]$: $H = (H_S, H_G, H_E, H_V, H_M, H_\pi)$ states whether humans define the initial problem or represented search space and, for every other component, whether they initiate, advise, approve, execute, or retain final decision authority. $H_S$ distinguishes authority over $S_0$ and later space expansion from candidate generation by $G$. The vector is descriptive, not numeric or ordered [Parasuraman et al., 2000].

**Faceted validator description.** The former labels $V_{\text{syntax}}$, $V_{\text{semantic}}$, $V_{\text{empirical}}$, $V_{\text{human}}$, and $V_{\text{formal}}$ mixed target properties, evidence sources, evaluator substrates, and decision roles; they were neither

Table 3: Predicate-level graph patterns. Every edge in an established witness must be implemented, feedback-dependent, closure-eligible, source-grounded, and connected in the listed order.

| Predicate | Required evidence condition | Required route and endpoint | Minimum recurrence and exclusions |
|---|---|---|---|
| Execution repair | Runtime-validation failure evidence | `artifact_revision` through $W_A$, then `feedback_control` to $E$ | A later execution; retry without a changed executable or protocol artifact and terminal review do not qualify |
| Experimental adaptation | Evaluation evidence that conditions a changed experimental action | Feedback-dependent `feedback_control` reaching $E$ | A later event explicitly typed as execution; selection alone, a fixed schedule, or code repair alone does not qualify |
| Artifact revision | Evaluation occurrence linked to an artifact | `artifact_revision` to a successor artifact through $W_A$ | A distinct versioned successor; formatting, copying, in-place mutation, and unlinked terminal assessment do not qualify |
| Discovery-cycle feedback | An evidence occurrence | `epistemic_update`, a feedback-controlled epistemic action, execution at $E$, then `observation` back to $V$ | A second distinct evidence occurrence; retry-only, append-only memory, and terminal-only paths do not qualify |

mutually exclusive layers nor a reliability order. Gate reliability evidence is recorded separately from runtime events and is called probabilistic calibration only when confidence–outcome correspondence is measured. Balanced accuracy, false-positive rate, mean score bias, and coder agreement are not calibration. The superseded binary fields remain reconstruction data only and are not inputs to the current schema.

**The validator as a reward channel.** Treating $V$ as distinct from the observation likelihood, Appendix A records three analytical separations. Posterior consistency belongs to the observation model and data-generating regime; a validator changes the inferential target only when it is coupled into a properly normalized update likelihood (or indirectly changes future designs); and a novelty component invisible to every available channel is non-identifiable. These observations diagnose where validator design can matter. They do not establish that any scalar validator measure is universally necessary or sufficient for trustworthy autonomous discovery.

## 5 Analysis of Core AI Scientist Systems

The frozen application contains six versioned cases from five lineages: Coscientist; AI Scientist 2024 and AI Scientist 2026 as two versions in one lineage; Agent Laboratory; Robot Scientist; and InternAgent (formerly NovelSeek). Each case is bounded by a reported configuration and task regime. The records are selected source-traceable examples, not "core" representatives of a population. They describe $S, G, E, V, M, \pi$, the boundary objects $O_{\text{env}}, A, W_A$, human authority, endpoints, events, artifacts, routed edges, reliability evaluations, and any closure witnesses that satisfy the Servo event–artifact contract.

Every cell is keyed by `case_id` and `predicate` in `servo2_closure_statuses.csv`. Established cells name their ordered `witness_ids`; each witness resolves event, edge, endpoint, artifact, evidence, source-location, and source-hash records in the release package.

The event graph preserves distinctions that a bare loop label would erase. In the AI Scientist lineage, pre-action literature filtering, runtime experiment or tree evaluation, manuscript assessment, and external review are separate events with different artifacts and destinations [Lu et al., 2024, 2026]. Coscientist and Agent Laboratory report execution or artifact-revision paths, but each closure predicate is stated only where an ordered within-case witness is available [Boiko et al., 2023, Schmidgall et al., 2025]. Robot Scientist reports a physical assay and model-update cycle within its bounded yeast-functional-genomics regime [King et al., 2004, Sparkes et al., 2010]; InternAgent reports multi-agent

| ID | Lineage | System/version | Configuration | Task regime |
|---|---|---|---|---|
| C01 | coscientist | Coscientist (2023 paper) | Reported demonstration | Six chemistry tasks |
| C02 | ai_scientist | AI Scientist 2024 (2024 paper) | Reported default | At most five computational experiments |
| C03 | ai_scientist | AI Scientist 2026 (2026 Nature) | Template-free tree | Computational tree search |
| C04 | agent_laboratory | Agent Laboratory (2025 paper) | Autonomous mode | Human-provided ML idea |
| C05 | robot_scientist | Robot Scientist (2010 review of Adam) | Adam yeast | Yeast functional genomics |
| C06 | novelseek | InternAgent (formerly NovelSeek) (2025 InternAgent) | Reported multi-agent | Twelve computational tasks |

Table 4: Six versioned cases from five lineages. IDs, versions, configurations, and task regimes are generated from the canonical Servo records.

| ID | Execution repair | Experimental adaptation | Artifact revision | Discovery-cycle feedback |
|---|---|---|---|---|
| C01 | {established} | {unknown} | {established} | {unknown} |
| C02 | {unknown} | {established} | {unknown} | {not established} |
| C03 | {unknown} | {established} | {unknown} | {unknown} |
| C04 | {established} | {unknown} | {established} | {not established} |
| C05 | {unknown} | {established} | {not applicable} | {established} |
| C06 | {established} | {established} | {established} | {not established} |

Table 5: Set-valued closure-predicate statuses. Columns are independent predicates, not levels of one closure scale; every established value requires an explicit within-case witness.

assessment, debugging, and evolution while Servo deliberately compresses agent-to-agent organization [InternAgent Team, Shanghai Artificial Intelligence Laboratory, 2025]. Human scope, manual execution, terminal authority, and external assessment remain separately recorded rather than being converted into a scalar autonomy or closure value.

The normative data model consists of the case, endpoint, artifact, event, edge, reliability-evaluation, closure-witness, closure-status, domain-anchor, and selection-ledger records named in `servo_schema.yaml`. The frozen 42-session, three-vendor multicoder audit used a superseded record-level rubric and did not code the current event, artifact, edge, or predicate units; it is retained only as a development audit.

### 5.1 Contrastive framework readings

The crosswalk in Table 2 is made concrete by three contrasts. First, a single loop label for AI Scientist (Nature 2026) hides the difference between stage and tree evaluators, a manuscript reviewer, and workshop review after manual submission selection. Second, Agent Laboratory's internal execution and artifact-revision paths are distinct from its paper-quality assessment and retained human authority. Third, InternAgent's native multi-agent architecture represents agent roles and communication more directly than Servo's system-level case record [InternAgent Team, Shanghai Artificial Intelligence Laboratory, 2025]. These are differences in foregrounded units, not a performance or superiority comparison.

The Robot Scientist is a negative case against an automatic claim of added diagnostic value: its source already exposes the relevant hypothesis, assay, update, and selection sequence [King et al., 2004, Sparkes et al., 2010]. Servo supplies a consistently validated record format and a documented mapping target, but does not thereby discover a hidden mechanism. The examples show only what the schema foregrounds.

### 5.2 Artifact Infrastructure: Agent-Native Research Artifacts

Liu et al. [2026] address a complementary problem—what the *output* of the loop should be and how to verify it. Agent-Native Research Artifacts (ARA) replace the narrative paper with a four-part ontology (`/logic`, `/src`, `/trace`, `/evidence`). In the final Servo facets, its Seal targets schema compliance, argumentative rigor, and reproducibility using deterministic checks, an automated Rigor

| ID | System/domain | Analysis target | Evaluator/evidence | Decision role | Fidelity boundary |
|---|---|---|---|---|---|
| DA01 | FunSearch / Formal mathematics | Program quality and task performance | Executable evaluator | Ranking and search control | Computational oracle |
| DA02 | AI Feynman / Physics | Symbolic fit to data | Symbolic-regression algorithm | Model search and fit | Computational data fit |
| DA03 | ChemCrow / Chemistry | Task performance and synthesis | Tools, execution, and assessment | Planning, execution, assessment | Computational and physical |
| DA04 | GNoME / Materials science | Phase stability and energy | Graph networks plus DFT | Filtering and active-learning update | Surrogate; separate experiment |
| DA05 | BioPlanner / Biology | Protocol planning and executability | Generator, evaluator, laboratory | Diagnosis and external execution | Protocol to physical laboratory |
| DA06 | Manning automated social science / Social science | Causal hypotheses and simulated effects | Causal model and LLM agents | Design and model update | Simulated subjects |
| DA07 | AutoML-Zero / Computer science | Evolved-algorithm benchmark performance | Executable benchmark | Selection and evolutionary update | Computational benchmark |

Table 6: Seven selected, frozen source anchors. The rows demonstrate how the vocabulary can be instantiated at named boundaries; they are not representative domain samples or a common outcome scale.

Auditor (82.6% detection), and sandboxed execution; significance and novelty remain assigned to human judgment [Liu et al., 2026]. Motivating figures: only 45.4% of reproduction requirements are fully specified and 90.2% of extension cost is failed exploration [Liu et al., 2026]. The `/trace` part externalizes $M_e/M_s$, but preserved traces both accelerated and anchored downstream agents—richer $M_e$ does not monotonically improve $\pi$.

# 6 Domain Analysis

Table 6 lists seven selected, frozen source anchors. Each row names one system and one bounded primary source; rows are not merged into a domain-level case, sampling frame, prevalence claim, or cross-domain outcome comparison.

The anchor rows support only source-local descriptions. They do not establish that a pattern holds across a domain, that one validator type causes an outcome, or that the seven anchors cover the relevant design space. DFT approximates physical energies rather than serving as a physical oracle, and predicted compounds still require synthesis and characterization [Cheetham and Seshadri, 2024].

# 7 Open Problems

The selected case and anchor sources motivate three prospective design questions. They do not establish prevalence or field-wide absence, and none of the proposed evaluations is executed here.

## 7.1 The Lack of Validated Automated Novelty Gates

The bounded sources used here do not report an independently validated automated novelty gate. This is a source-limited evidence statement, not a finding that every automated novelty score is unreliable or that no other system supplies such evidence. The cited failures concern adjacent properties: paper-quality bias, hallucinated tables, proof correctness, structural violations, or review overlap. They motivate caution but do not measure novelty discrimination. Novelty, correctness, significance, and aggregate review quality remain separate endpoints.

*Prospective validation design requirements:* These are necessary design requirements, not a sufficient checklist for declaring a gate validated. A preregistration must define the target population, temporal cutoff, domain strata, novelty prevalence or sampling scheme, primary discrimination endpoint (for example sensitivity/specificity or AUC), minimally relevant performance, and confidence-interval width. A prospective power or precision calculation must then determine benchmark-item and rater counts per domain; "three domains" and "five experts" may serve as diversity floors but never substitute for that calculation. Experts label independently, human–human agreement is reported with uncertainty, and held-out adjudicated field-novelty labels use explicit prior-art search kept separate from significance and correctness. Follow-up duration, censoring, and the rule for later priority or prior-art outcomes must also be preregistered. The automated system is evaluated against held-out adjudicated labels rather than the majority vote from the same raters. High agreement with a small panel or separation of published from withheld findings remains only an initial benchmark result. A recent benchmark takes a step by restricting to post-2024 findings to control contamination [Liu et al., 2025], though it evaluates inspiration-based decomposition rather than satisfying this design.

Novelty, significance, and correctness are distinct properties that the bounded sources may score separately and then combine at a decision point (The AI Scientist rates ideas on separate interestingness, feasibility and novelty scales, yet uses an aggregate `Overall` paper score [Lu et al., 2024]). Functional separation of generation from reviewing, reflection, or ranking does not imply statistical or organizational independence. The selected sources do not report independent, contamination-controlled evidence that such a channel discriminates field novelty.

### 7.2 The BED–Practice Gap for $\pi$

Equation 1 specifies a one-step, myopic information criterion. The bounded LLM-case sources used here do not state that their policies optimize EIG or a sequential information objective. Modern BED can use variational estimators, amortized policies, and explicit probabilistic surrogates [Foster et al., 2019, 2021]; BED-LLM [Choudhury et al., 2026] provides a separate example for conversational queries. The open question is integration with a reported $G$–$\pi$ action interface and scientific evidence path, not an asserted field-wide absence.

*Prospective success criterion:* A deployed AI Scientist narrows the survey gap if it maintains an observation-updated probabilistic model—exact, variational, ensemble, or other documented surrogate—and uses a validated EIG or sequential information objective to select actions $a_t$ under explicit budgets. Evaluation should compare discovery-relevant utility and information efficiency against matched heuristic policies. Direct Bayesian representation of open-ended natural-language or program hypotheses is a separate, stronger representation problem; it is not required to count a practical BED integration.

Architecturally, a system may maintain an explicit probabilistic surrogate or use variational and amortized EIG estimation [Foster et al., 2019, 2021]. What remains difficult in open-ended $\mathcal{S}$ is defining and updating a representation on which those objectives are meaningful. EIG-based selection presupposes an already represented hypothesis space, whereas how genuinely new hypotheses or representations *arise* is a separate generative problem [Yanai and Lercher, 2019].

### 7.3 The Multi-Fidelity Integration Gap

Test-time compute has created a structural asymmetry: hypothesis generation is now fast and cheap, while experiment execution $E$ remains physically rate-limited. Principled criteria for selecting among costly information sources already exist in multi-fidelity Bayesian optimization: continuous-fidelity models trade approximation cost against target information [Kandasamy et al., 2017], and cost-sensitive mutual-information acquisition selects among simulations and real evaluations [Song et al., 2019]. The gap identified here is therefore not the absence of mathematical escalation criteria. It is their unvalidated integration into the surveyed open-ended AI Scientist workflows, where fidelity changes may alter protocols, outcome semantics, and physical feasibility.

The selected anchors expose several proxy-to-physical boundaries. GNoME predicts 380,000 stable compounds via DFT [Merchant et al., 2023]; 736 match structures independently present in or added to the ICSD, including 184 added after the project began. These are concurrent records by others rather than syntheses the system triggered. ChemCrow reports wet-lab execution, whereas physical characterization remains protocol-specific [Bran et al., 2024, Gromski et al., 2019]. BioPlanner generates protocol artifacts without thereby establishing a biological-validity feedback witness [O'Donoghue et al., 2023]. These anchors motivate, but do not establish the prevalence of, a multi-fidelity integration problem.

This design problem is distinct from Op1 and Op2: a proposed implementation would need to make evidence comparable across fidelities and expose fidelity and cost to $\pi$. Existing MFBO supplies acquisition criteria under stated surrogate assumptions; whether those assumptions and protocol transitions hold in an AI Scientist workflow is an empirical question.

Autonomous laboratories such as A-Lab [Szymanski et al., 2023], ORGANA [Darvish et al., 2024], and AutoLabs [Panapitiya et al., 2025] show that execution automation and fidelity selection are separate engineering questions. They are cited as adjacent examples, not as an exhaustive audit of multi-fidelity policy evaluation.

*Prospective success criterion:* A system closes the integration gap if it implements a documented multi-fidelity acquisition rule, including cost and cross-fidelity uncertainty, and prospectively outperforms matched fixed-fidelity or fixed-escalation baselines on experimentally validated discoveries per unit physical cost. This tests adoption and calibration in an AI Scientist workflow rather than claiming to invent the underlying criterion.

## 8 Discussion

**Validator design as a diagnostic vocabulary.** Unreliable validation can distort what a system records as progress—an ecosystem-scale symptom is the documented surge in fabricated citations [Zhao et al., 2026], and strong task performance need not imply an internalized world model [Vafa et al., 2025]. The earlier gate-reliability/"trustworthy closure" association is withdrawn as conceptually circular, not merely left untested. The framework instead asks property-specific questions: which channel fires when, where its output is routed, what property it assesses, what bias or calibration evidence exists, at what experimental fidelity, and with what external independence? Any future outcome analysis must use gate-independent external endpoints $Y_q$.

**Human oversight.** The categorical authority vector makes explicit where humans initiate, advise, approve, execute, or retain final authority; it is not a quantity to minimize. Human authority in novelty/significance assessment and policy is appropriate for the surveyed workflows because their automated proxies do not yet provide independently validated judgments of those targets. This is a current-workflow finding, not a claim that those judgments are non-automatizable in principle [Parasuraman et al., 2000].

**Towards collective AI co-scientists.** Some analyzed systems, including InternAgent, are explicitly multi-agent [InternAgent Team, Shanghai Artificial Intelligence Laboratory, 2025]; this paper nevertheless collapses each implementation into one system-level $(S, G, E, V, M, \pi)$ tuple and does not model agent-level roles, communication, disagreement, or authority. The relevant extension is therefore from that system-level abstraction to an agent graph, shared-memory process, collective validator, or Dec-POMDP-like model, rather than the introduction of multi-agent systems as such. Collective Predictive Coding [Taniguchi et al., 2025] and ARA [Liu et al., 2026] offer possible starting points for shared epistemic state. Such an extension must preserve diversity: AI reviewers already overlap more than independent humans [Kim et al., 2026], risking the transient diversity that benefits epistemic communities [Zollman, 2010].

**Limitations.** Servo is non-exhaustive and abstracts away agent-level and social organization [Taniguchi et al., 2025]. The six cases informed framework development, while the seven anchors are partial records rather than complete held-out graphs. No untouched case has yet been instantiated end-to-end under a prospectively frozen contract. The present evidence therefore supports internal consistency and a loss-aware standards mapping, not demonstrated out-of-sample portability; a future holdout that triggers a schema change must be reclassified as formative. A predicate status concerns only its bounded case and witness and does not imply scientific outcome quality. Servo supplies neither a causal model nor a generative law for missing routes, costs, fidelity transitions, or artifacts. The frozen 42-session, three-vendor multicoder audit predates the current event–artifact contract and is retained only as a historical development audit; its agreement coefficients do not establish reproducibility, source entailment, construct validity, human reliability, real-world utility, or comparative superiority.

## 9 Conclusion

We introduced Servo $(S, G, E, V, M, \pi)$ as a non-exhaustive, source-traceable event-and-evidence schema. Its six versioned cases from five lineages and seven frozen anchors illustrate how to record evaluation events, artifact versions, routes, authority, and four conservative closure predicates. The schema reuses generic provenance concepts while adding scientific target, reliability, authority, epistemic-route, and closure-witness semantics. These formative cases and partial anchors do not establish out-of-sample portability, prevalence, causality, comparative superiority, or scientific outcome effects.

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## Conference For AI Scientists 2026 - AI Involvement Checklist

This checklist documents the role of AI systems at each stage of this research. The checklist has two parts: a **Research Stage Assessment** (items 1–4) rating AI involvement and iteration effort, and an **AI System Documentation** section (items 5–6).

### Research Stage Assessment

1. **Hypothesis development.**

   Answer: **[B]**

   Iteration Effort: **[Medium]**

   Explanation: The core Servo framework tuple $(S, G, E, V, M, \pi)$, the original $V$-completeness thesis (now retained only as correction history), the human-authority representation, and the three open problems were conceived by the human author. A frontier LLM agent elaborated framework components, identified literature-level instantiations, and proposed theoretical connections. The human author directed all conceptual decisions and made several substantive corrections to the AI's elaborations.

2. **Experimental design and implementation.**

   Answer: **[B]**

   Iteration Effort: **[Medium]**

   Explanation: This submission contains no computational experiments. The methodology for cross-system analysis (system selection criteria, component extraction protocol, domain comparison structure) was designed by the human author. A frontier LLM agent executed literature retrieval, extracted framework components from primary sources, and performed citation-level fact verification across four verification sweeps.

3. **Analysis of data and interpretation of results.**

   Answer: **[C]**

   Iteration Effort: **[Medium]**

   Explanation: The LLM agent extracted Servo components from primary sources and performed citation-level fact verification. The human author reviewed and corrected the annotations. The six case records and seven source anchors are an author-directed synthesis with source provenance; there is no independent human gold standard, and historical agreement statistics are not presented as evidence of reliability or construct validity for the current schema.

4. **Writing.**

   Answer: **[C]**

   Iteration Effort: **[High]**

   Explanation: The LLM agent drafted the majority of the manuscript text through multiple revision cycles; the human author provided key conceptual passages (framework definitions in §4, core open problems in §7), directed structural decisions, reviewed all drafts, and made final editorial judgments.

### AI System Documentation

5. **AI systems used.**

   Description: A frontier large language model with code execution and long-document processing capabilities was used as the primary research and writing assistant. The system operated as an interactive CLI agent with file-system access, able to read primary sources, execute and debug Python simulations, and iteratively revise LaTeX manuscripts.

6. **Observed AI Limitations.**

   Description: (1) Systematic citation fabrication: the agent generated plausible but non-existent quotes and statistics, requiring four dedicated citation-verification sweeps using parallel sub-agents. (2) Analytical overconfidence: the agent initially overclaimed empirical significance for mechanically-entailed theoretical consequences; multiple adversarial review rounds were needed to correct framing and remove overclaims. (3) Context drift:

over long sessions, the agent occasionally regressed previously corrected errors, requiring explicit re-verification of integrity items before each commit.

## Conference For AI Scientists 2026 - Reproducibility and Responsibility Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state the non-exhaustive Servo schema contribution, its six bounded cases and seven frozen anchors, and the prospective open questions. Appendix A supplies bounded analytical observations rather than a new theorem, while the case and anchor records are explicitly non-representative source annotations.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Limitations are discussed in §8 (framework abstracts domain cost structures, social layer, non-stationarity) and throughout the domain analysis (§6).

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete proof?

   Answer: [NA]

   Justification: The paper does not claim a new theorem or proof. Appendix A states three bounded analytical observations, identifies the extra conditions needed to invoke standard results on posterior consistency and model misspecification, and expressly leaves adaptive open-ended regimes outside their scope. The selected source anchors illustrate vocabulary use rather than serving as a proof.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all information needed to reproduce the main experimental results?

   Answer: [Yes]

   Justification: This paper contains no computational experiment, but its main case annotations and closure predicates depend on the Servo event–artifact package. The synchronized package, validator, canonical records, generated tables, evidence ledger, audit record, and corrected manuscript are bound in the immutable release at `https://github.com/ysys143/servo-caisc2026/releases/tag/servo-corrected-v3.0.4`. Its manifest and attestation provide file-level checksums for the revised status matrix.

5. **Open access to data and code**

   Question: Does the paper provide open access to data and code?

   Answer: [Yes]

   Justification: The release above binds the current case, endpoint, artifact, event, edge, reliability, closure-witness, closure-status, domain-anchor, selection-ledger, contract, code, tables, audit record, and corrected manuscript files.

6. **Experimental setting/details**

   Question: Does the paper specify all details necessary to understand the results?

   Answer: [Yes]

   Justification: The cross-domain analysis methodology is described in §5 and §6. No computational experiments were performed in this submission.

7. **Experiment statistical significance**

   Question: Does the paper report error bars or statistical significance information?

   Answer: [NA]

Justification: This paper contains no computational experiments and reports no statistical results requiring error bars or significance tests.

8. **Experiments compute resources**

   Question: Does the paper provide sufficient information on compute resources?

   Answer: [NA]

   Justification: This submission contains no computational experiments. The meta-analysis requires only standard document processing and no specialized compute.

9. **Research Ethics**

   Question: Does the research conform with the NeurIPS Code of Ethics, except where CAISc's AI authorship policies apply?

   Answer: [Yes]

   Justification: This is a meta-analysis of published systems with no human subjects, no proprietary data, and no safety risks. AI authorship is documented per CAISc policy.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive and negative societal impacts?

    Answer: [Yes]

    Justification: §8 addresses both the potential for accelerating scientific discovery and the risk of premature convergence under widespread AI Scientist deployment (particularly via the multi-fidelity integration gap and the current lack of validated automated novelty gates).

# A Analytical Separation of Observation, Reward, and Novelty Channels

We separate three objects that are easily conflated: the *observation likelihood* $p(o \mid \theta, d)$ used for Bayesian updating, the context-dependent *validator* $V(z)$ used as a reward or acceptance signal, and any additional channel that carries evidence about novelty or significance. Observation 2 studies only the restricted special case $V(o)$ with all other context held fixed so that Berk's fixed-i.i.d. result can apply; it is not a general model of novelty or acceptance. The observations below apply standard results to fixed idealized settings. They are not new general theorems for adaptive experimental design or open-ended science.

**Observation 1: posterior consistency belongs to the observation channel.** In a dominated, fixed i.i.d. model $\{p_\theta : \theta \in \Theta\}$ with true distribution $p_{\theta^\star}$, posterior consistency can follow when the model identifies the truth, the prior puts positive mass on every Kullback–Leibler neighborhood of it, and suitable consistent tests separate it from complements of its neighborhoods, together with the regularity conditions needed by the relevant consistency theorem [Ghosal and van der Vaart, 2017]. Compactness, identification, and prior support stated only in a topological sense are not substitutes for those conditions. The cited result therefore does not establish consistency for the adaptive, non-stationary design sequence of an AI Scientist without an additional theorem for that regime.

Equation 1 defines a one-step information objective. Maximizing it is myopically optimal for that objective by construction, but this does not establish global optimality for a sequential scientific policy. Let $\mathcal{H}_{t-1}$ contain the past designs and observations, let $d_t$ be measurable with respect to that history, and define $\mathrm{EIG}_t = I(\theta; O_t \mid \mathcal{H}_{t-1}, d_t)$. Posterior concentration does not require a divergent sum of these one-step values. For example, when $\theta$ is discrete with finite entropy, the conditional-information chain rule gives

$$\sum_{t \geq 1} \mathbb{E}[\mathrm{EIG}_t] = I(\theta; \mathcal{H}_\infty) \leq H(\theta), \tag{6}$$

so finite cumulative expected information is compatible with posterior concentration. If $V$ remains outside the likelihood, it has no direct role in this fixed-model consistency statement, although policy use of $V$ can indirectly change which future designs and observations occur.

**Observation 2: a coupled validator changes the pseudo-true target only through a normalized likelihood.** For a fixed design $d$, suppose an update uses

$$q_{\theta,d}(o) = \frac{p(o \mid \theta, d)\exp\{\beta V(o)\}}{Z_\theta(d)}, \qquad Z_\theta(d) = \int p(u \mid \theta, d)\exp\{\beta V(u)\}\,du < \infty. \tag{7}$$

Let $F$ denote the fixed i.i.d. data-generating distribution. Under the regularity, integrability, separation, and prior-mass conditions required for a Berk-type misspecification result, posterior mass is driven toward a minimizer set rather than automatically toward a unique point:

$$\Theta^\dagger = \arg\min_{\theta\in\Theta} \mathrm{KL}(F \,\|\, q_{\theta,d}) \tag{8}$$

Only when this set is a singleton do we denote its member by $\theta^\dagger$ [Berk, 1966]. Validator-induced drift means that this minimizer set differs from the corresponding projection under $p(\cdot \mid \theta, d)$; an outcome-dependent $V$ can alter the normalizer $Z_\theta(d)$ even when $V$ has no explicit $\theta$ argument, but non-constancy alone neither guarantees nor is required to prove a target change. An additive constant in $V$ cancels between numerator and normalizer, while $\beta = 0$ gives $q_{\theta,d} = p(\cdot \mid \theta, d)$. A validator used only by the policy may still alter future design selection, but that is an adaptive-design effect, not the fixed-i.i.d. conclusion of Berk [1966].

**Observation 3: novelty invisible to every available channel remains unidentified.** Write the latent state as $\theta = (\theta_{\text{nov}}, \psi)$ and suppose the prior factorizes as $\Pi(d\theta_{\text{nov}}, d\psi) = \Pi_{\text{nov}}(d\theta_{\text{nov}})\Pi_\psi(d\psi)$. If the joint likelihood of all observed channels depends on $\psi$ but not on $\theta_{\text{nov}}$, then the posterior marginal of $\theta_{\text{nov}}$ equals $\Pi_{\text{nov}}$ for every dataset. This full factorization from every likelihood-relevant latent component is essential: independence only from already identified components would not exclude indirect updating through a prior-correlated nuisance variable. The novelty component is therefore not consistently identifiable from those channels. Identification requires an additional informative channel, but the premise does not imply that the channel must be human in principle. The survey instead supports the narrower empirical statement that novelty and significance judgments in the systems examined here remain human-mediated or lack a validated automated gate.

These observations support the paper's diagnostic separation of observation, reward validation, and novelty judgment. They do not prove that the empirical $V$-completeness ordinal is necessary for trustworthy autonomous discovery, establish causal effects of validator design, or settle the paper's open-ended and adaptive setting. The former scalar-completeness and gate-reliability/"trustworthy closure" hypotheses are withdrawn, not retained as provisional hypotheses; any future outcome study requires property-specific, gate-independent endpoints.

# B Cross-System Coding Reliability

The Servo event–artifact schema is the current normative contract. It records six versioned cases from five lineages, their endpoints, artifacts, events, edges, reliability evaluations, predicate-specific witnesses and statuses, seven frozen source anchors, and a selection ledger. The four closure predicates are set-valued and bounded by case, version, configuration, and task regime; no single stored loop label is authoritative. The records are author-interpreted source annotations. Structural validation cannot establish semantic entailment, construct validity, or scientific outcomes.

Three independent, model-pinned LLM coders from different vendors coded the earlier 14-system rubric (13 systems had all three usable outputs). The resulting Fleiss statistics—historical `Vcalibrated` $\kappa$=0.79, loop status $\kappa$=0.74, semantic-validation presence $\kappa$=1.0, mean $\kappa$=0.68, and holistic $V$-completeness $\kappa$=0.39—apply only to those superseded coding fields. They are retained solely as automated audit history and are not transferred to the revised constructs.

The subsequent two-vendor recode over 14 systems produced raw agreement 13/14 and Cohen's $\kappa$=0.86 for the historical present-versus-gating distinction. That exercise documents one post-hoc construct refinement, chiefly the AI Scientist (Nature 2026) reviewer-role ambiguity; it is not a reliability study of the final faceted taxonomy. The legacy $V_{\text{present}}$, $V_{\text{gating}}$, and $V_{\text{calibrated}}$ columns therefore remain in the supplement only for reconstruction.

The frozen 42-session, three-vendor multicoder audit used a then-current manual to code 14 source records. It is retained only as a historical development audit. Its record-level token unions, Jaccard summaries, MASI-based Krippendorff $\alpha$, and disagreement log neither align current events and edges nor reproduce the current closure witnesses. No statistic from this audit is transferred to the current contract or case statuses. Its provenance is retained in the revision-history appendix.

## C Per-Domain Analysis

This appendix comments on the same seven frozen anchors as Table 6. Each paragraph is bounded to one named system and source. It does not describe a domain population.

**FunSearch.** FunSearch [Romera-Paredes et al., 2024] records program generation with executable scoring in a bounded computational task. This anchor does not supply field-novelty evidence.

**AI Feynman.** AI Feynman [Udrescu and Tegmark, 2020] evaluates equation candidates against problem-local data. The anchor does not establish cross-task memory transfer.

**ChemCrow.** ChemCrow [Bran et al., 2024] combines tool use with reported chemical execution. Wet-lab characterization remains protocol- and artifact-specific.

**GNoME.** GNoME [Merchant et al., 2023] routes DFT relaxation and model uncertainty into computational search for phase stability. DFT approximates physical energies rather than serving as a physical oracle: 736 predicted structures match ICSD entries, including 184 added after the project began, while synthesis and functional characterization remain separate [Merchant et al., 2023, Cheetham and Seshadri, 2024].

**BioPlanner.** BioPlanner [O'Donoghue et al., 2023] generates protocol artifacts, but protocol text alone does not establish a biological-validity feedback witness.

**Manning automated social science.** The selected workflow [Manning et al., 2024] uses statistical tests over generated experimental designs. This one source is not evidence about automated social science generally.

**AutoML-Zero.** AutoML-Zero [Real et al., 2020] routes executable benchmark results into search. Benchmark performance does not establish interpretability.

**Cross-anchor reading.** The rows show only that the same vocabulary can be instantiated at seven named source boundaries. They do not support aggregation, causal comparison, or domain-level generalization.

## D Post-Submission Revisions

This manuscript continues to be refined after the CAISc 2026 submission deadline. This appendix is the running record of every difference between it and the archival version of record, so that the two can be reconciled entry by entry. Each entry states what changed, why, and whether the version of record is thereby wrong or merely less precise.

**R1. Leiden Declaration signatory count (Section 7).** *Status in the version of record: unsupported by the source cited for it.* The version of record states that the Leiden Declaration was "signed by over 130 mathematicians," attributing that figure to the declaration document. The document states no signatory count at all: its only numerals are section numbers and the year of the meeting that produced it. The count is maintained on the separate signature site, which recorded 3,071 signatories when checked on 20 July 2026. The defect is therefore not that the figure was understated but that it was attributed to a text that does not contain it. This revision cites the signature site for the count, marks it as of a date because it grows over time, and notes that the declaration itself gives none. The substantive claim the sentence supports—that AI-produced proofs may carry errors that resist detection without expert scrutiny—is stated in the declaration and is unaffected.

**R2. Formal validators and RL stability (Section 6).** *Status in the version of record: overclaim; the sentence asserts more than the evidence supports.* The version of record states that formal mathematics has "a binary $V_{\text{formal}}$ that makes RL stable and the loop fully closed." A deterministic pass/fail oracle settles whether a candidate proof or program is acceptable, but the stability of policy learning also depends on reward density, the state space, the policy update rule, the exploration strategy, and function approximation; the surveyed formal systems do not isolate the validator as the cause of stable training (DeepSeek-Prover-V1.5 additionally uses supervised fine-tuning and an MCTS variant with intrinsic rewards, and FunSearch couples the evaluator to an LLM and an evolutionary procedure). The clause is replaced here with "enables mechanically closed evaluation within the formalized task," matching the wording already used for the same systems in Appendix C. No other claim depends on the removed clause.

**R3. Scoring of novelty, significance and correctness (Section 7).** *Status in the version of record: contradicted by the cited source.* The version of record cited "The AI Scientist's 1–10 score" as an instance of collapsing the three properties into one scalar. The cited system in fact rates ideas on *separate* interestingness, feasibility and novelty scales, and its reviewer emits several axes; the collapse occurs not in scoring but at the accept/reject decision, which thresholds a single aggregate. The sentence now locates the collapse where it happens. The open problem is unchanged.

**R4. "Greedy" as a coding, not a source claim (Section 7).** *Status in the version of record: unsupported interpretation.* The cited source describes archive-conditioned idea generation but defines no argmax, best-score or greedy selection rule; "greedy" is this paper's own coding. The sentence now says so.

**R5. Two different quantities compared (Section 5).** *Status in the version of record: contradicted in metric meaning.* The version of record read "balanced accuracy 0.65 vs. human 0.66," which invites reading 0.65 as agreement with human reviewers. It is the balanced accuracy of predicting ICLR 2022 accept/reject labels; 0.66 is human inter-reviewer consistency measured on a different pool. The two are now distinguished.

**R6. Class definition and Agent Laboratory (Introduction).** *Status in the version of record: contradicted by the cited source.* The opening sentence defines the class as agents that *independently* generate hypotheses and cited Agent Laboratory among them; in that system the research idea is supplied by a human. The citation has been removed from the class-defining list. Agent Laboratory remains analysed throughout the paper.

**R7. Robot Scientist table cells (Table 4).** *Status in the version of record: one unsupported, one mislabelled.* The novelty-measure cell read "Stat. test"; the statistical test in the cited source adjudicates growth phenotypes and gene-function hypotheses, not scientific novelty, so the cell now reads "None (corr.)." The policy cell "Active learn." is not supported by the 2010 review used for that column; the caption now attributes the active-learning policy to the primary source that documents it.

**R8. What "calibration" denotes (Section 4).** *Status in the version of record: a term used in a weaker sense than the word normally carries, without saying so.* The version of record calls a validator "uncalibrated" on the strength of discrimination statistics. Calibration and discrimination are independent properties: balanced accuracy, F1 and AUC measure how well a scorer separates classes, while calibration—whether stated confidence matches empirical frequency—is measured by reliability curves, expected calibration error or Brier score. No surveyed source reports the latter, so neither "calibrated" nor "uncalibrated" is established in the strong sense. The paper's own coding protocol already defines the field operationally as *not systematically biased* rather than as probabilistically calibrated. This revision states that definition in the body where the term is first made operative. The construct, its coding, and the reported inter-coder agreement are unchanged; only the term's scope is now explicit.

**R9. The automated reviewer is not a gate (Section 5).** *Status in the version of record: contradicted by the cited source.* The version of record described the automated reviewer as the system's "internal acceptance gate." In the cited source the reviewer scores finished manuscripts and configurations; what advances the search is a stage evaluator, a best-first policy and a compute budget, and the manuscripts sent for external review were chosen by hand. The reviewer is an evaluation instrument, not a gate. The sentence now says so, and the acceptance bias it does exhibit is stated with the figure that supports it (false-positive rate 0.45 against 0.17 for human reviewers).

**R10. Coding corrections and their effect on reported agreement (Appendix B).** *Status in the version of record: reference labels contained demonstrable errors, and one reported statistic depended on them.* An audit of the released coding sheet against the paper's own protocol found eleven cells that the protocol, the primary sources, or the manuscript itself contradict. `Vsyntax` was coded 0 for every system, including five that execute code, against a rubric whose example of the field is "code compiles/executes"; the blind coders assigned 1 unanimously on four of them. `Vhuman` was coded 1 for AI Scientist (Nature 2026) on the strength of an external post-hoc workshop review, which the protocol's "required for the validity/novelty signal" test does not cover, and 0 for Agent Laboratory, contradicting this paper's own table, which records that system's novelty measure as human-delegated. `Vcompleteness` was 3 for AI Scientist (Nature 2026), a value the scale licenses only when the top automated layer is calibrated; that flag had been corrected to 0 earlier without propagating to the ordinal. The remaining ordinals were re-derived from the scale anchor after the `Vsyntax` correction rather than copied from the coders; that derivation independently reproduces the coder consensus in every case. *Effect on reported statistics:* the Fleiss agreements among the three blind coders, which are the figures this paper quotes, are identical before and after the correction, as are the pairwise coder agreements. What moves is agreement between the coders and the author's reference labels—necessarily, because those labels were revised. The released reliability report now states both the original and the corrected values and warns that the rise is an artefact of the revision rather than evidence of reliability.

**R11. InternAgent / InternAgent citation identity (Table 4).** *Status in the version of record: bibliographic entry does not match the work as it now stands.* The entry keyed `zhang2025novelseek` gave the title and individual author list under which arXiv:2505.16938 first circulated. Later versions of the same preprint (v3, the one audited here) retitle the work *InternAgent* and place it under a laboratory team byline. The citation is to the right work—the identifier is unchanged—but a reader retrieving 2505.16938 today finds a different title and author line than the bibliography gives. The entry now carries the current title and byline, records the former name, and keeps the key so that earlier citations of this work still resolve. The manuscript continues to call the system InternAgent, which is how it is named in the analysis released with this paper, and notes the retitling in the caption of Table 4, where the system is introduced.

**R12. What the 736 GNoME matches are (Section 7).** *Status in the version of record: the phrase is the source's own, but it invites a reading the source does not support.* The version of record wrote that of 380,000 predicted stable compounds "only 736 have been independently experimentally realized," which is verbatim from the cited abstract. The cited Methods make the mechanism explicit: the 736 are matches between GNoME's predicted structures and entries already present in, or added to, the ICSD when it was queried in January 2023, and 184 of them are new since the project began. They are therefore structures other groups made independently or concurrently, not syntheses the system prompted. Used as the numerator of a realization rate, the original phrasing suggests a predict-then-synthesize pipeline that did not operate. The sentence now states the mechanism. This strengthens rather than weakens the point it supports—that escalation to physical work is decided outside the system—since the matches were not triggered by any escalation criterion at all.

**R13. Epistemic status of the formal claims (Abstract; Section 4; Appendix A).** *Status in the version of record: less precise and materially overstates the scope of the mathematical contribution.* The version of record presents three numbered propositions, a bundled assumption, and proof sketches as if they established results for a tractable AI Scientist loop. The paper's purpose is a diagnostic framework, however, and the cited statistical results do not supply a new theorem for adaptive, non-stationary, open-ended search. This revision replaces that theorem-and-proof posture with three explicitly bounded analytical observations, removes the unused theorem environments and their cross-references, and states in the abstract, background, checklist, conclusion, and domain discussion that the appendix imports standard fixed-setting distinctions rather than proving the framework. The six-component framework, three-way closure distinction, and provisional survey claims are unchanged; their evidential status is made explicit rather than upgraded by mathematical presentation.

**R14. Posterior consistency and cumulative EIG (Appendix A).** *Status in the version of record: partly incorrect.* The version of record treats compactness, identification, prior support, and a bespoke design-separation clause as a standard sufficient condition for posterior consistency under Ghosal and van der Vaart [2017], without stating the Kullback–Leibler prior-mass and testing conditions used by the relevant classical results or proving an adaptive-design analogue. Its proof sketch also says that positive but summable one-step EIG leaves the posterior short of a point mass. That claim is false in general: for a discrete parameter with finite entropy, the conditional-information chain rule bounds total expected EIG by initial entropy even when the posterior concentrates. The revision states the fixed-i.i.d. conditions only schematically and with their scope, removes the unsupported adaptive conclusion, and describes Equation 1 as a one-step myopic criterion rather than a globally optimal scientific policy. This narrows the theoretical rhetoric without weakening the observation-versus-reward distinction, the empirical BED–practice gap, or its falsification test.

**R15. Misspecified validator coupling and KL projection (Appendix A).** *Status in the version of record: mathematically under-specified and broader than the cited result.* The version of record writes the coupled update only up to proportionality, invokes Berk [1966] without restricting the claim to its fixed-i.i.d. misspecification setting, assumes a unique KL projection, and gives "outcome dependence" as if it were a sufficient drift condition. The revision defines the finite normalizer $Z_\theta(d)$, fixes the design and data-generating distribution, states the needed regularity and prior-mass qualifications, and uses the KL-minimizer set $\Theta^\dagger$ unless uniqueness is separately available. Drift now means that this set differs from the projection under the original likelihood; an additive constant cancels and an outcome-dependent validator need not change the target. Policy-only effects are separated as adaptive design selection, not attributed to Berk's theorem. The correction preserves the paper's diagnostic warning about coupling while withdrawing any claim that the surveyed systems instantiate this asymptotic result.

**R16. Novelty non-identification and the human channel (Section 7; Appendix A).** *Status in the version of record: the non-identification premise is correct, but its human-only conclusion exceeds that premise.* If every observed likelihood is independent of an a priori independent novelty component, its posterior marginal remains the prior; those channels cannot identify it. What follows is the need for an additional informative channel, not a theorem that this channel must always be human or that human validation is irreplaceable in principle. The revision makes that logical boundary explicit and limits the empirical conclusion to the surveyed systems: their novelty and significance judgments are human-mediated or lack a validated automated gate. This leaves the novelty open problem and its falsification criterion unchanged while removing a universal impossibility claim.

**R17. Table 4 reconciled with the coding sheet.** *Status in the version of record: two rows disagreed with the released coding they summarise.* The novelty row read "None" for Co-scientist, where the sheet codes a human gate, and "Human-deleg." for Agent Laboratory, where the sheet codes a biased automated gate—the latter also contradicting Section 5, which describes that system's reviewer as biased. The $H$ row rendered Coscientist's 0.75 as "$\approx$1" while giving Agent Laboratory's 0.8 exactly, so one table rounded two neighbouring values by different rules. Both rows now reproduce `novelty_gate` and `A_H` verbatim and the row is renamed to name what it reports. This also settles the Robot Scientist cell left open by R7: the sheet codes a predefined novelty criterion, which is what the cell now reads.

**R18. Residual claim boundaries and analytical premises (Sections 4 and 7; Appendix A).** *Status in the version of record: partly incorrect, and the R13–R16 revision remained incomplete.* The version of record identifies a sequential policy with a one-step EIG maximizer, treats validator reliability and human review too categorically, and states the novelty posterior-invariance argument without excluding prior coupling through other likelihood-relevant latent variables. The first mathematical revision narrowed these claims but left several stronger English sentences in place and retained the insufficient independence premise. This revision makes the EIG rule explicitly myopic, factorizes the novelty prior from every likelihood-relevant latent component, limits human mediation to a finding about the surveyed workflows, and recasts cross-domain validator patterns as provisional associations rather than necessity or causality. The Korean source is narrowed to the same scope. These changes repair the logical propagation of R13–R16 without changing the six-component framework, the three open problems, or their proposed falsification tests.

**R19. Correction status and gate semantics (Abstract; Table 4; Appendix B).** *Status in the version of record: materially inaccurate in several claims, while the earlier post-submission citation and coding language remained insufficiently precise.* Appendix D records corrections to factual, coding, statistical, and mathematical statements, so readers should not be directed to the archival text alone when relying on a corrected claim. The notice now asks readers to pair the version-of-record citation with this dated correction and states that the correction should be linked from the official record when possible. Table 4 also called a biased automated score a "novelty gate" for AI Scientist (Nature 2026), although the paper and released coding set `V_gating=0`: the reviewer scores completed outputs but does not advance search or select external submissions. The row is therefore renamed "Novelty assessment," the non-gating status is explicit, and the caption distinguishes the legacy `novelty_gate` field from operational gating. Finally, the abstract withdraws the claim of direction consistency: because the six-system outcome coding partly overlaps $V$, even the direction of association remains untested. These corrections preserve the framework and hypotheses while removing unsupported evidential status.

**R20. Validator roles, closure scope, and novelty validation (Sections 3–8; Table 6).** *Status in the version of record: partly incorrect and conceptually under-specified.* Related Work described AI Scientist (Nature 2026)'s biased automated reviewer as the gating layer even though the system description and R19 establish that it is a non-gating assessment instrument. The text now treats $V_{\text{gating}} = 0$ only as failure to establish a calibrated decisive gate and states that the binary field does not distinguish absence, non-gating, unidentified, and identified-but-uncalibrated cases. The manuscript also withdraws the proposed reliability partial order and any substantive use of the low-agreement holistic $V$-completeness ordinal, replacing it with a vector of gate identity, target property, bias/calibration evidence, experimental fidelity, and external independence. The introductory servo analogy is restricted to trustworthy acceptance feedback, and formal mathematics in Table 6 is described as exact and mechanically closed only for formal validity, not as a perfect scientific oracle or a fully closed discovery process. Finally, the novelty "falsification test" is recast as a prospective validation design requiring independent expert labels, human–human agreement, prior-art adjudication, separate endpoints, held-out evaluation, uncertainty reporting, and later priority checks. These changes preserve the six-component vocabulary and the open problems but remove a scalar validator hierarchy, overbroad closure language, and an inadequate novelty ground-truth claim.

**R21. Interfaces and loop evidence (Sections 4, 5, and 7).** *Status in the version of record: partly incorrect and materially under-specified.* The version of record assigned both $G$ and $\pi$ the output type $h$ and then placed them in an operational sequence that was not a function composition. The revision makes $G$ generate a candidate set, makes $\pi$ select a typed action $(h, d, P)$, and states the resulting update loop explicitly. It also replaces unsupported decimal human-intervention scores with a categorical component-wise authority vector. For AI Scientist (2024), the earlier table's computational-closure label is retained but its rationale is corrected: the primary system description returns execution failures and metrics to the experiment agent for conditional replanning, whereas the terminal simulated reviewer is a separate assessment layer and is not what closes that internal loop. The table and released coding now record this internal empirical-feedback channel without treating the reviewer as an operational gate. The Nature 2026 system is analyzed separately: its performance, training-dynamics, plot-quality, and stage-evaluator signals control best-first expansion and stage transitions, whereas its paper reviewer is terminal assessment; calibration of the search-controlling signals is not established. Finally, the BED criterion now admits exact, variational, ensemble, and other documented probabilistic surrogates, while direct Bayesian representation of natural-language hypotheses is identified as a stronger separate problem; the former experiment-fidelity problem is narrowed to a multi-fidelity integration gap in light of existing cost–information acquisition criteria. These corrections preserve the six-component vocabulary and the AI Scientist (2024) computational-loop classification, but repair the component interfaces, the evidence for that classification, the human-authority coding, and the claimed novelty and success criteria of two open problems.

**R22. Validator context and scope (Sections 4 and 8).** *Status in the version of record: partly incorrect and materially under-specified.* The version of record typed the validator as $V : o \to r$, which can represent observation scoring but not novelty, significance, reproducibility, or acceptance judgments that depend on a candidate claim, protocol, memory, external prior art, and an evaluation-time cutoff. The revised interface is $V : z \to r$, where $z$ exposes those contextual inputs and $r$ may contain separate property-specific outputs; the operational loop now constructs $z_t$ explicitly. The restricted $V(o)$ used in Appendix A is identified as a special fixed-context case rather than a general novelty model. The manuscript also no longer claims a complete POMDP grounding: it distinguishes the candidate search space $S$ from a POMDP world-state space $\mathcal{X}$, lists the kernels a full instantiation would require, and describes POMDP belief and policy concepts as organizing terminology because those kernels and a reward model are not specified here. The novelty open problem is narrowed from a performance conclusion to the documented absence of independently validated novelty-specific gates; paper quality, hallucination, correctness, and review agreement are retained only as adjacent general assessment failures, not novelty-discrimination measurements. Finally, cross-domain relation language is removed, and the multi-agent limitation now states that systems such as InternAgent are multi-agent but are collapsed into one system-level tuple. These changes preserve the component taxonomy and open research questions while correcting their formal expressiveness, evidential status, and abstraction boundary.

**R23. Observation kernel and final coding facets (Sections 4 and B).** *Status in the version of record: partly incorrect and materially under-specified.* The version of record made $E$ a deterministic map from hypothesis and protocol to one observation even though its likelihood–reward distinction and BED discussion require the selected design to alter an observation distribution. The revised executor is the design-conditioned kernel $E(\mathrm{d}o \mid h, d, P, x)$, with deterministic execution as a Dirac special case, and the operational loop now samples from that kernel. This locates the observation model inside the six-component tuple without adding a seventh component and states explicitly that EIG and belief updates are not computable when a system does not expose the required probability law. The former syntax/semantic/empirical/human/formal layer labels also mixed target property, evidence source, evaluator substrate, decision role, and external independence. They are replaced in substantive analysis and Table 4 by those separate facets; legacy layer columns remain only for audit reconstruction. Likewise, absence of documented bias is no longer called calibration. Gate reliability evidence is reported by evidence type, and probabilistic calibration is reserved for an evaluated confidence–outcome correspondence. Finally, all reported $\kappa$ values are explicitly limited to the earlier rubric: they do not validate the R20–R23 facets, contextual validator, human-authority vector, or diagnostic utility. A frozen-manual, prospective independent recoding remained future evaluation at R23 and is now reported in R24 without the abandoned free-form baseline comparison. These changes preserve the six component names while repairing the observation path, validator taxonomy, statistical terminology, and reproducibility claim.

**R24. Frozen faceted recoding audit (Limitations and Appendix B).** *Status in the version of record: materially under-specified and not applicable to the revised taxonomy.* The version of record reported agreement for an earlier mixed-field rubric and therefore supplied no reproducibility evidence for the final validator facets. The revision freezes hash-bound local-source packets, the final coding manual and schema, a 42-cell schedule, strict provenance validation, isolated host-login CLI execution, and four adversarial smoke gates. Three vendors then produced one accepted fresh-session coding for each of 14 source records: six manuscript core cases and eight supplementary scope cases. All 42 accepted outputs and their hashes are sealed before analysis. Facet-wise mean pairwise Jaccard values are 0.567–0.706 and MASI-based Krippendorff $\alpha$ values are 0.165–0.338; 90 record–facet disagreements are retained without overwriting raw outputs. These small-$n$ quantities compare record-level unions of facet tokens, not alignment among complete validator channels. They are descriptive audit results, not pass thresholds, inferential validation, a human reference standard, or evidence that Servo outperforms a free-form or competing framework. The change strengthens traceability and demonstrates application of the final schema while leaving diagnostic superiority and downstream utility untested.

**R25. Channel-aware core cases and construct crosswalk (Sections 3, 5, and Appendix B).** *Status in the version of record: materially under-specified and partly conflates validator presence with operational feedback.* The version of record compressed each system's heterogeneous assessment mechanisms into layer and loop labels, did not preserve which target, evidence source, evaluator, decision role, and feedback path belonged to the same channel, and offered no construct-level comparison with adjacent frameworks. After the R24 outputs were frozen, the six core cases were therefore adjudicated against the same hash-bound local-source packets. The released channel ledger resolves every evidence ID to an exact quote, page, and PDF hash; the separate component sheet records $S$, $G$, $E$, $M$, $\pi$, component-specific human authority, and experimental fidelity. The resulting table distinguishes operational feedback from terminal and external assessment and supersedes the record-level majority projection for substantive case reading. A framework-native crosswalk and three contrastive readings identify Servo's selected system-level resolution while acknowledging InternAgent's advantage for agent organization and the Robot Scientist as a negative case where the central closed loop was already explicit. The change improves traceability and construct clarity but does not establish population reliability, universal superiority, exhaustiveness, causal utility, or improved system design.

**R26. Framework boundary corrections (Sections 3, 4, 6, and Appendix B).** *Status in the version of record: partly incorrect and materially under-specified.* The version of record claimed that no shared cross-system framework existed, although direct AI-Scientist surveys already organize autonomous discovery by capabilities, mechanisms, and workflow stages. The revision therefore compares Servo first with the unified frameworks of Wei et al. and Tie et al.; POMDP, InternAgent, CPC-MS, and ARA remain complementary adjacent formalisms rather than selected direct baselines. Servo's contribution is narrowed to making selected system-level distinctions explicit and comparable, not supplying the first shared vocabulary or a universally superior taxonomy. The substantive domain table and appendix also replace residual syntax/semantic/empirical/human/formal layer shorthand and common reliability rankings with target property, evidence and evaluator, decision role and feedback, fidelity boundary, and local bottleneck fields. The six-case validator table is now explicitly an author-interpreted source synthesis: channel split/merge and bounded "None identified" rules are disclosed, and neither R24's record-level token agreement nor the later adjudication is presented as independent channel-level reproducibility. Finally, R23's single stochastic $E$ mixed a system-controlled executor with the environment's observation law. The present formulation keeps $E_{\text{exec}}$ as the system component and introduces $O_{\text{env}}$ as an auxiliary environment/measurement model outside the six-component tuple. These corrections leave the six component names and the three open problems intact, while narrowing the originality and reproducibility claims and clarifying the framework's system–environment boundary.

**R27. Event channels and open-ended policy interfaces (Sections 2, 4, 7, and Appendix B).** *Status in the version of record: partly incorrect and materially under-specified.* The version of record, and revisions through R26, represented validation as one post-observation call even though the substantive cases distinguish pre-action filters, in-execution evaluators, terminal review, and external assessment. The revised $V$ is a family of event-indexed channels with a trigger phase and routed destination; the operational loop preserves pre-action and later events instead of collapsing them into one reward vector. The coding manual now uses trigger phase as a split criterion, but this is an author-specified refinement of the frozen R24 material, not a new independent recoding. The earlier gate-reliability/"trustworthy closure" association is withdrawn rather than merely called untested because its outcome definition contained gate reliability; future studies require gate-independent, property-specific endpoints. The generator now returns an explicit search-space expansion, and the policy selects fidelity under a cost function and remaining budget. Finally, the novelty-gate proposal is relabeled as necessary design requirements and requires prospective power or precision calculations, prespecified prevalence and discrimination targets, confidence-interval precision, and follow-up/censoring rules. These changes make the six-component vocabulary capable of locating the stated open problems but do not establish open-endedness, cost-aware optimization, external outcome validity, or a sufficient novelty-validation protocol in any surveyed system.

**R28. Closure coding, audit status, and search-space authority (Sections 4, 5, 7, and Appendix B).** *Status in the version of record: partly incorrect and materially under-specified.* The version of record and subsequent tables coded Coscientist and Agent Laboratory as "partial" even after the operational criterion defined any implemented feedback edge to later generation, policy, memory, stage, or represented search space as computational closure. Source-grounded channels show such edges in both systems, so the current table codes them computationally closed within their demonstrated task or revision workflows. Human problem selection, manual execution, terminal authority, and incomplete novelty or physical validation remain separate authority, fidelity, and evidential limitations. R24 is also reclassified as a historical development audit: its three-vendor record-level facets preceded the final trigger, routed channel, space-expansion, fidelity, cost, budget, and $H_S$ fields and therefore do not establish reproducibility of the current framework. The final six-case channel table remains the author's source-grounded synthesis. The novelty discussion now distinguishes functional generator/reviewer separation, which several systems implement, from independently validated and contamination-controlled novelty discrimination, which the surveyed reports do not establish. Finally, the authority vector adds $H_S$ so human control of the initial problem and represented search space is not silently folded into $G$. These changes preserve the diagnostic framework and open problems while correcting two case labels and narrowing its reproducibility and architecture claims.

**R29. Canonical schema, derived closure, and projection integrity (Sections 4 and 5, Tables 4 and 5, and Appendix B).** *Status in the version of record: materially under-specified, with later revisions internally inconsistent.* The version of record did not provide a machine-checkable source of truth for the framework, and revisions through R28 still distributed current judgments among legacy fields, free-text component sheets, split channel/timing files, manually stored bilingual display strings, and prose. This allowed corrected closure definitions to coexist with stale loop labels and let Appendix B restate a hypothesis already withdrawn as definitionally circular. The current revision freezes Servo schema 1.0, joins each event channel's trigger, destination, facets, evidence and source hash in one canonical row, records all six authority fields and system-level expansion/fidelity/cost/budget states, and derives computational closure from implemented routed edges rather than a stored label. A fail-closed validator checks evidence ownership, channel compatibility, historical/current separation, claim mapping and bilingual projections before generating both tables. The mixed `systems.csv`, compatibility exports, and R24 outputs remain available for reconstruction but are not current semantic inputs. This consolidation improves internal consistency and auditability; it is not a new independent recoding, source-entailment study, construct-validity result, or test of Servo's superiority or scientific-outcome prediction.

**R30. Portable event–evidence contract and bounded case identity (Abstract; Sections 4, 5, and 6; Appendix B).** *Status in the version of record: materially under-specified; the post-submission Schema 1 interpretation was also too coarse.* The version of record and revisions through R29 treated a system record and a routed validator channel as sufficient current units and collapsed heterogeneous feedback into one computational-closure judgment. They did not require version, configuration, and task-regime identity along a witness; distinguish runtime events from reliability studies; or represent versioned artifacts and artifact revision explicitly. Schema 2.0 preserves the six-component core while placing $O_{\text{env}}$, artifact state $A$, and optional artifact synthesis or revision $W_A$ at its boundary. It defines an event–artifact graph and five set-valued predicates—execution repair, experimental adaptation, artifact revision, discovery-cycle feedback, and human-mediated feedback—with conservative source-traceable witnesses. The application is restated as six versioned cases from five lineages and seven selected frozen source anchors. This change improves identity, provenance, and falsifiability of the diagnostic descriptions; it does not validate the schema, establish scientific outcomes, estimate a population, infer causes, or show comparative superiority.

**R31. Local reproducibility-package status (Reproducibility and Responsibility Checklist; Appendix B).** *Status in the version of record: absent, and later post-submission wording overstated public availability.* The archival record did not contain Schema 2.0. A complete working-tree package now exists locally, including the normative schema and case, endpoint, artifact, event, edge, reliability, closure-witness, closure-status, domain-anchor, and selection-ledger records. No public repository snapshot or DOI for that complete package has yet been verified, and it is neither published nor part of the official record. The reproducibility and open-access checklist answers are therefore changed to "No—not yet." This correction separates local preparation from public availability; it does not change the paper's diagnostic argument or supply external reproduction, peer validation, or official erratum linkage.

**R32. Predicate conformance and provenance. Scope:** Related Work; §4, §5, §8; reproducibility checklist. *Status in the version of record: absent; the earlier post-submission contract specified a full pattern only for discovery-cycle feedback and did not separate source silence from negative evidence.* This revision defines minimum graph patterns for all five predicates, adds machine-readable decision bases for established, unknown, not-established, and not-applicable statuses, and reclassifies source-silent negatives as unknown. It maps base objects and relations to PROV-DM and Workflow Run RO-Crate while limiting Servo's increment to scientific diagnostic semantics. It also withdraws demonstrated portability: the six complete cases were formative and the seven anchors are partial, so no prospective held-out full instantiation has yet been reported. The public 2.0.0 snapshot predates these changes; the checklist remains "No—not yet" for the current analysis until a synchronized successor release is published.

**R33. Closure-axis correction and cross-representation conformance. Scope:** Abstract; §4; §5; Appendix B; public analysis package. *Status before this correction: the actor providing feedback was mixed with path topology, and the AI Scientist 2026 adaptation cell omitted a source-stated execution transition.* Human mediation is now an actor facet rather than a fifth closure predicate. The four remaining predicates describe path topology. Event classes and edge types are constrained to compatible six-component actors and source–destination pairs. The AI Scientist 2026 experimental-adaptation status is corrected to established because metric- and plot-guided node selection is followed by child-code generation and reported parallel execution. This does not promote that path to discovery-cycle feedback, which still requires an explicit epistemic update and a second evidence occurrence.

**R34. Explicit execution occurrences in adaptation witnesses. Scope:** Predicate contract and public analysis package. The experimental-adaptation validator now requires a later event whose class is explicitly `execution`; reaching endpoint $E$ or repeating an evaluation event is insufficient. The AI Scientist 2024 and InternAgent witnesses now name their already source-grounded execution events directly. This tightens conformance without changing their established statuses.

**R35. Portability-claim boundary. Scope:** Related Work; §4; normative contract and provenance crosswalk. The manuscript no longer labels the current contract or event–artifact graph portable. The evidence establishes an operational, machine-readable representation and a construct-level mapping to PROV-DM and Workflow Run RO-Crate. It does not establish executable serialization, standards conformance, or out-of-sample portability; those require a tested serializer and a prospectively frozen holdout application.

**R36. Explicit tuple–graph admissibility maps. Scope:** §4; normative contract and validator. The manuscript now states the complete event-class–actor map $\eta$ and edge-type–component-transition map $\tau$ enforced by the validator. This makes the six-component tuple and event–artifact graph jointly checkable while preserving the distinction between an invalid mapping and a source that does not report one.

**R37. Actor–topology independence and explicit predicate patterns. Scope:** §4; normative contract, validator, and regression suite. Human mediation may now qualify a structurally valid discovery-cycle path without being mistaken for a closure type or an automation claim. Table 3 states the required evidence condition, ordered route, endpoint or recurrence, and exclusions for all four predicates; the validator continues to reject retry-only, append-only, disconnected, and otherwise nonconforming paths.

**Scope of revision.** The entries above are the complete set of changes made since the archival deposit, R1–R37. Entries R3–R12 narrow or correct statements that the citation audit judged unsupported by, contradicted by, or less precise than their cited sources. Entries R13–R16 record the subsequent adversarial mathematical review: they replace the proposition/assumption/proof-sketch presentation with bounded analytical observations, correct the EIG statement, delimit the conditions under which the Ghosal–van der Vaart and Berk references may be invoked, and distinguish mathematical non-identification from the surveyed use of human judgment. R17 reconciles two table rows with the released coding sheet. R18 records the adversarial recheck that completed the propagation of the earlier mathematical corrections, including the stronger prior-factorization premise and the removal of residual universal, causal, and human-only wording. R19 corrects the citation guidance, distinguishes novelty assessment from operational gating, and withdraws the untestable direction-consistency statement. R20 makes gate-zero semantics explicit, withdraws the holistic validator order from substantive analysis, narrows closure and formal-oracle language, and replaces the novelty majority-agreement threshold with a prospective validation design. R21 types the generator–policy interface, identifies the internal experimental feedback that supports the AI Scientist (2024) computational-loop coding, replaces scalar human-intervention scores, and reframes the BED and proxy-to-physical criteria around accepted surrogate BED and multi-fidelity integration methods. R22 makes validation context-dependent, limits the POMDP relation to organizing concepts rather than a complete instantiation, separates the absence of reported novelty validation from adjacent assessment failures, removes residual cross-domain association language, and states the system-level abstraction of multi-agent cases. R23 places a stochastic, design-conditioned observation kernel inside $E$, replaces mixed validator layers with explicit facets, reserves calibration for measured probability–outcome correspondence, and limits the old agreement statistics to their pre-revision rubric. R24 replaces the abandoned free-form baseline comparison with an evidence-bound audit of six core and eight supplementary records, freezes 42 independent three-vendor codings, and reports facet disagreements and agreement coefficients only as descriptive development-audit diagnostics. R25 replaces the record-level majority projection as substantive evidence with a source-adjudicated channel ledger, restores the complete component and authority fields for the six cases, and adds a framework-native construct crosswalk with contrastive and negative cases. R26 adds the closest AI-Scientist survey frameworks to the direct comparison, propagates the final validator facets through the domain analysis, marks the six-case channel table as an author interpretation rather than an independently reproduced result, and separates the system executor from the auxiliary environment observation law. R27 replaces the single post-observation validator call with routed event channels, withdraws the circular gate-reliability/closure hypothesis, makes search-space expansion and budgeted fidelity explicit, and turns the novelty benchmark checklist into precision-driven design requirements. R28 applies the closure rule consistently to Coscientist and Agent Laboratory, demotes R24 to a pre-final-schema development audit, separates functional reviewer architecture from validated novelty discrimination, and adds authority over the search space to the human-authority vector. R29 consolidates the current six-case semantics in a versioned schema, derives closure and bilingual tables from canonical records, rejects legacy/R24 inputs from current projections, and corrects the residual Appendix-B restatement of the withdrawn closure hypothesis. R30 supersedes Schema 1 as the current interpretation: it introduces bounded versioned case identity, explicit artifact state and revision, typed event and edge records, and five predicate-specific closure judgments with ordered witnesses; it also replaces domain aggregation with seven selected frozen source anchors. R31 records the then-local package state; R32 records the later public 2.0.0 snapshot, clarifies all predicate and evidential-status contracts, adds the provenance mapping, and marks the revised working tree as awaiting a synchronized successor release. The six component labels persist, as do the three prospective design questions, but their interfaces and operationalization have changed materially: Schema 2 adds bounded identity, versioned artifacts, $W_A$, typed event and edge records, and predicate-specific witnesses in place of Schema 1's channel and single-closure interpretation. The application remains non-representative source annotation. In addition to the entry corrected in R11, R21 adds Parasuraman et al. on function-specific human–automation authority and Kandasamy et al. and Song et al. on multi-fidelity Bayesian optimization; R26 adds Wei et al. and Tie et al. as the closest direct AI-Scientist survey frameworks.

**Current-interpretation supersession index.** R23's stochastic executor formulation is read with R26's executor/environment separation; R24's agreement results are read only as the development audit delimited by R28 and R29; in particular, R24's contemporaneous phrases "final coding manual and schema" and "application of the final schema" are superseded and do not describe schema 1.0; R25's six-case synthesis is read through the canonical event records introduced in R29; and R27's event-channel interface and R29's Schema 1 closure derivation are superseded by the Schema 2 event–artifact graph, bounded case identity, and predicate-specific witness contract introduced in R30. References in earlier entries to "final" Schema 1 tables or current six-case closure labels are historical statements, not present semantics. This index does not alter the historical text or chronology of those entries.